Message Queue Lab — Submission

Exercise 1 — RabbitMQ: Crash Before ACK

☐ Terminal output from the **unsafe** mode — both the crash run and the restart, showing `applied` and `business_effect_count`

```
# 2110521-software-arch/message-broker on main ?:2 with changes in .venv
> python rabbitmq/demo1_crash_ack.py publish --mode unsafe --run unsafe1
2026-10-07T09:26:34.616+00:00 CONFIRMED id=c98a8f87-1250-4650-b80b-2afb57b9fcf1 payload={'action': 'create_order'}
# 2110521-software-arch/message-broker on main ?:2 with changes in .venv
> python rabbitmq/demo1_crash_ack.py worker --mode unsafe --run unsafe1 --crash-before-ack
2026-10-07T09:26:43.936+00:00 READY queue=lab.failure.crash.unsafe.unsafe1; processes one delivery then exits
2026-10-07T09:26:43.936+00:00 RECEIVED id=c98a8f87-1250-4650-b80b-2afb57b9fcf1 tag=1 redelivered=False
2026-10-07T09:26:43.965+00:00 DB COMMITTED: applied=True, business_effect_count=1
2026-10-07T09:26:43.965+00:00 INTENTIONAL CRASH: database committed, ACK NOT SENT (exit code 99)
# 2110521-software-arch/message-broker on main ?:2 with changes in .venv
> python rabbitmq/demo1_crash_ack.py worker --mode unsafe --run unsafe1 --crash-before-ack
2026-10-07T09:26:53.562+00:00 READY queue=lab.failure.crash.unsafe.unsafe1; processes one delivery then exits
2026-10-07T09:26:53.562+00:00 RECEIVED id=c98a8f87-1250-4650-b80b-2afb57b9fcf1 tag=1 redelivered=True
2026-10-07T09:26:53.591+00:00 DB COMMITTED: applied=True, business_effect_count=2
2026-10-07T09:26:53.591+00:00 INTENTIONAL CRASH: database committed, ACK NOT SENT (exit code 99)
# 2110521-software-arch/message-broker on main ?:2 with changes in .venv
> python rabbitmq/demo1_crash_ack.py worker --mode unsafe --run unsafe1
2026-10-07T09:27:07.442+00:00 READY queue=lab.failure.crash.unsafe.unsafe1; processes one delivery then exits
2026-10-07T09:27:07.442+00:00 RECEIVED id=c98a8f87-1250-4650-b80b-2afb57b9fcf1 tag=1 redelivered=True
2026-10-07T09:27:07.472+00:00 DB COMMITTED: applied=True, business_effect_count=3
2026-10-07T09:27:07.472+00:00 ACK SENT
```

☐ Terminal output from the **safe** mode — both the crash run and the restart, showing `applied` and `business_effect_count`

```
# 2110521-software-arch/message-broker on main ?:2 with changes in .venv
> python rabbitmq/demo1_crash_ack.py publish --mode safe --run safe1
python rabbitmq/demo1_crash_ack.py worker --mode safe --run safe1 --crash-before-ack
python rabbitmq/demo1_crash_ack.py worker --mode safe --run safe1
2026-10-07T09:27:19.787+00:00 CONFIRMED id=717aa68d-8586-48f2-8946-d810a1aaca89 payload={'action': 'create_order'}
2026-10-07T09:27:19.940+00:00 READY queue=lab.failure.crash.safe.safe1; processes one delivery then exits
2026-10-07T09:27:19.940+00:00 RECEIVED id=717aa68d-8586-48f2-8946-d810a1aaca89 tag=1 redelivered=False
2026-10-07T09:27:19.970+00:00 DB COMMITTED: applied=True, business_effect_count=1
2026-10-07T09:27:19.970+00:00 INTENTIONAL CRASH: database committed, ACK NOT SENT (exit code 99)
2026-10-07T09:27:20.108+00:00 READY queue=lab.failure.crash.safe.safe1; processes one delivery then exits
2026-10-07T09:27:20.108+00:00 RECEIVED id=717aa68d-8586-48f2-8946-d810a1aaca89 tag=1 redelivered=True
2026-10-07T09:27:20.108+00:00 DB COMMITTED: applied=False, business_effect_count=1
2026-10-07T09:27:20.108+00:00 ACK SENT
```

☐ Written explanation (3–5 sentences): Why does the unsafe mode produce a duplicate? How does the safe mode prevent it, and what role does the single transaction play?

RabbitMQ will try to send the message to consumer until it receives at least 1 ACK. The consumer in this context polls for 1 message from message queue, process and send ACK back. In unsafe mode, the consumer doesn't check whether the table has the requested row or not and always do INSERT. When the unsafe consumer doesn't send ACK back to broker, the broker will repeatedly try to send message, which cause consumer to perform duplicated effects. In safe mode, the consumer checks whether the table has the requested before execution. Therefore, even if the consumer fails to send ACK and receives duplicated request, there is no duplicated effect.

Exercise 2 — RabbitMQ: Late Subscribers

☐ Terminal output from the **durable** consumer showing both `BEFORE` and `AFTER` received

```
# 2110521-software-arch/message-broker on feat/message-broker in .venv
> python rabbitmq/demo2_late_subscriber.py durable --run fanout1
2026-10-11T10:34:41.438+00:00 READY durable queue=lab.failure.fanout.fanout1.saved; C
trl+C to stop
2026-10-11T10:34:41.438+00:00 DURABLE RECEIVED id=01267bad-6508-4b4f-a050-26d521f74cb
1 body={"event_id": "01267bad-6508-4b4f-a050-26d521f74cb1", "order_id": "order-demo",
 "sequence": 1, "payload": {"text": "BEFORE"}}
2026-10-11T10:35:03.507+00:00 DURABLE RECEIVED id=df342b3a-090f-4a37-9c63-4c761de3942
a body={"event_id": "df342b3a-090f-4a37-9c63-4c761de3942a", "order_id": "order-demo",
 "sequence": 1, "payload": {"text": "AFTER"}}
```

- ☐ Terminal output from the **temporary** consumer showing `AFTER` received and `BEFORE` missing

```
# 2110521-software-arch/message-broker on feat/message-broker in .venv
> python rabbitmq/demo2_late_subscriber.py temporary --run fanout1
2026-10-11T10:34:57.218+00:00 READY temporary queue=amq.gen-wLXWPX6xMvnvul9HLlpT6g; C
trl+C to stop
2026-10-11T10:35:03.507+00:00 TEMPORARY RECEIVED id=df342b3a-090f-4a37-9c63-4c761de39
42a body={"event_id": "df342b3a-090f-4a37-9c63-4c761de3942a", "order_id": "order-demo
", "sequence": 1, "payload": {"text": "AFTER"}}
```

- ☐ Written explanation (3–5 sentences): Why did the temporary queue miss `BEFORE`? What would need to change for it to receive `BEFORE`?

The durable queue didn't miss `BEFORE` because it is bounded to the broker before `BEFORE` was even published when `setup` is run. So, after the broker routes message to durable queue, the durable queue stores the message until durable queue's consumer wakes up and consume `BEFORE`. The temporary queue missed `BEFORE` because it is bounded to broker after `BEFORE` was published, so the consumer of temporary queue only receives `AFTER`. To make temporary queue receives `BEFORE`, we need to bound the queue with broker before message is published.

Exercise 3 — RabbitMQ: Delivery vs. Completion Order

- ☐ Terminal output from the **3-worker** run showing completion order

```
2026-10-11T10:56:09.480+00:00 W1 DONE A elapsed=5.01s
^C2026-10-11T10:56:25.923+00:00 Stopping consumer
# 2110521-software-arch/message-broker on feat/message-broker U:1 ?:2 with changes in
 .venv
> python rabbitmq/demo3_completion_order.py worker --run order2 --name W1
2026-10-11T10:56:28.612+00:00 READY W1 queue=lab.failure.order.order2; Ctrl+C to stop
2026-10-11T10:56:34.278+00:00 W1 START A delay=5s redelivered=False
2026-10-11T10:56:39.279+00:00 W1 DONE A elapsed=5.00s
```

message-broker — python rabbitmq/demo3_com...

```
2026-10-11T10:56:05.478+00:00 W2 DONE B elapsed=1.00s
^C2026-10-11T10:56:26.545+00:00 Stopping consumer
# 2110521-software-arch/message-broker on feat/message-broker U:1 ?:2 with changes in
 .venv
> python rabbitmq/demo3_completion_order.py worker --run order2 --name W2
2026-10-11T10:56:30.320+00:00 READY W2 queue=lab.failure.order.order2; Ctrl+C to stop
2026-10-11T10:56:34.280+00:00 W2 START B delay=1s redelivered=False
2026-10-11T10:56:35.280+00:00 W2 DONE B elapsed=1.00s
```

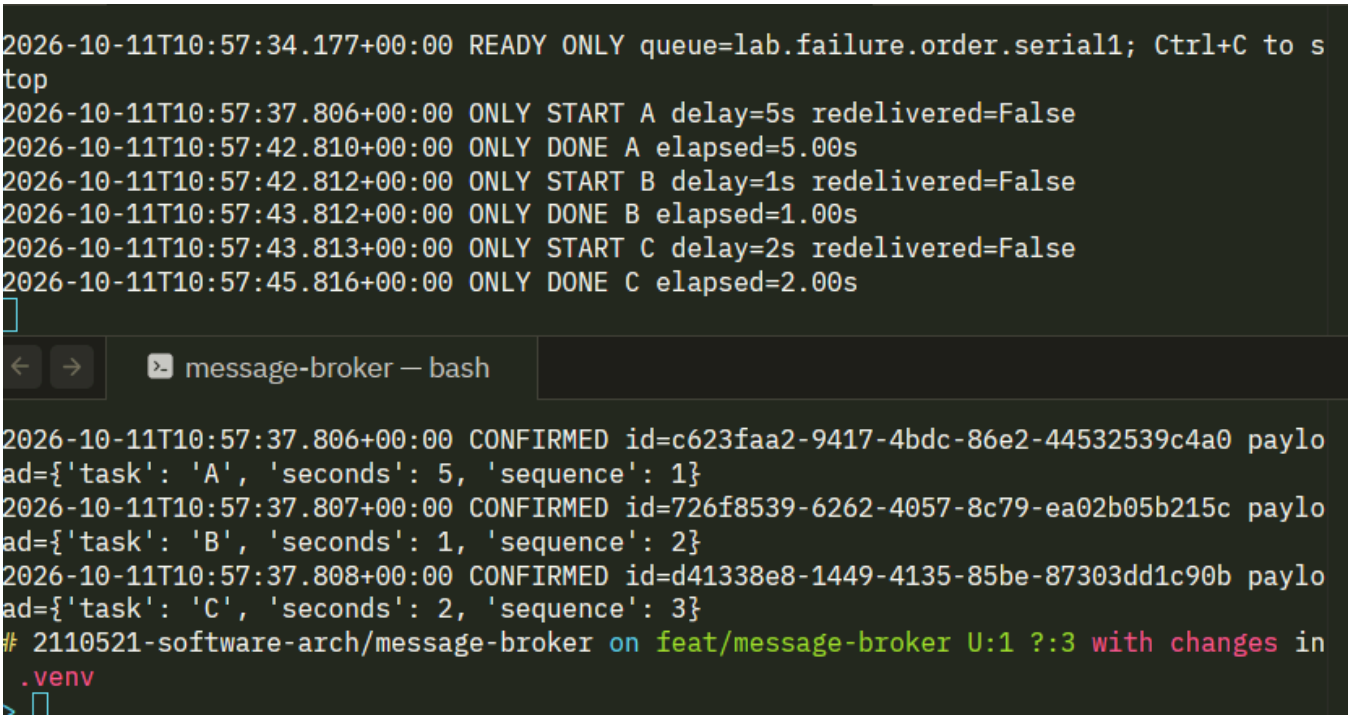
message-broker — python rabbitmq/demo3_com...

```
2026-10-11T10:56:06.480+00:00 W3 DONE C elapsed=2.00s
^C2026-10-11T10:56:27.158+00:00 Stopping consumer
# 2110521-software-arch/message-broker on feat/message-broker U:1 ?:2 with changes in
 .venv
> python rabbitmq/demo3_completion_order.py worker --run order2 --name W3
2026-10-11T10:56:31.816+00:00 READY W3 queue=lab.failure.order.order2; Ctrl+C to stop
2026-10-11T10:56:34.281+00:00 W3 START C delay=2s redelivered=False
2026-10-11T10:56:36.283+00:00 W3 DONE C elapsed=2.00s
```

message-broker — bash

```
2026-10-11T10:56:34.278+00:00 CONFIRMED id=6be174c0-8455-4f80-828f-7788a2aa2bf5 paylo
ad={'task': 'A', 'seconds': 5, 'sequence': 1}
2026-10-11T10:56:34.280+00:00 CONFIRMED id=8021c321-b361-4b6a-82ff-5a3c64cadf70 paylo
ad={'task': 'B', 'seconds': 1, 'sequence': 2}
2026-10-11T10:56:34.281+00:00 CONFIRMED id=bc299393-b1ef-4992-b1a1-3d6f23cbaa67 paylo
ad={'task': 'C', 'seconds': 2, 'sequence': 3}
# 2110521-software-arch/message-broker on feat/message-broker U:1 ?:2 with changes in
 .venv
```

□ Terminal output from the **1-worker** run showing completion order



- [ ] Written explanation (3–5 sentences): Why do the two runs produce different completion orders? When would you choose one worker over three?

3-worker run has completion order of B -> C -> A which is different to message order because tasks are spread out and processed concurrenry. 1-worker run has completion order of A -> B -> C which is the same to message order because task are sent to 1 worker and done in sequential manner. We might choose 1-worker in case that the system must respect in-order processing.

Exercise 4 — Kafka: Consumer Groups and Partitions

- [ ] Evidence logs (or terminal screenshots) showing partition assignment for group sizes **2, 4, and 5**

Group of 2

Publisher


```
{"timestamp": "2026-10-11T15:12:18.863871+00:00", "kind": "published", "event_id": "43498796-409f-4574-b5f2-dc0b531abc49",
"topic": "lab.ex4.two1", "partition": 3, "offset": 0}
{"timestamp": "2026-10-11T15:12:18.864037+00:00", "kind": "published", "event_id": "7a37bbec-11a9-4f54-9250-12a99f4e3e75",
"topic": "lab.ex4.two1", "partition": 3, "offset": 1}
{"timestamp": "2026-10-11T15:12:18.864146+00:00", "kind": "published", "event_id": "5ce185c6-073e-4cf5-8267-98ee50c016cd",
"topic": "lab.ex4.two1", "partition": 3, "offset": 2}
{"timestamp": "2026-10-11T15:12:18.864231+00:00", "kind": "published", "event_id": "b8fbac61-b440-4a5a-8bd8-f92719ecf700",
"topic": "lab.ex4.two1", "partition": 3, "offset": 3}
{"timestamp": "2026-10-11T15:12:18.864306+00:00", "kind": "published", "event_id": "cd74db69-3ee6-4788-85a9-4bf7d02b0921",
"topic": "lab.ex4.two1", "partition": 3, "offset": 4}
{"timestamp": "2026-10-11T15:12:18.864396+00:00", "kind": "published", "event_id": "8a294563-1fc2-4cc2-b278-ab1256b82835",
"topic": "lab.ex4.two1", "partition": 3, "offset": 5}
{"timestamp": "2026-10-11T15:12:18.864485+00:00", "kind": "published", "event_id": "3e9489d0-81fe-4e04-85b2-4e2dea7ec670",
"topic": "lab.ex4.two1", "partition": 3, "offset": 6}
{"timestamp": "2026-10-11T15:12:18.864559+00:00", "kind": "published", "event_id": "805a7b15-8faa-406d-85e1-da2c1ae430f8",
"topic": "lab.ex4.two1", "partition": 3, "offset": 7}
{"timestamp": "2026-10-11T15:12:18.864633+00:00", "kind": "published", "event_id": "83e6cb73-1664-40fc-8f63-88d79f1adef0",
"topic": "lab.ex4.two1", "partition": 3, "offset": 8}
{"timestamp": "2026-10-11T15:12:18.864701+00:00", "kind": "published", "event_id": "23c005c9-ac4f-4173-ac7c-2416e1c39b12",
"topic": "lab.ex4.two1", "partition": 3, "offset": 9}
{"timestamp": "2026-10-11T15:12:18.865712+00:00", "kind": "published", "event_id": "a4ccec2b-9d54-4ff3-abc9-0ede4cccc5c5",
"topic": "lab.ex4.two1", "partition": 0, "offset": 0}
{"timestamp": "2026-10-11T15:12:18.865930+00:00", "kind": "published", "event_id": "4541e3e4-1637-4294-9117-6f424c41c6ac",
"topic": "lab.ex4.two1", "partition": 0, "offset": 1}
{"timestamp": "2026-10-11T15:12:18.866040+00:00", "kind": "published", "event_id": "027c3037-0946-4bcc-845e-4cbee46b48e5",
"topic": "lab.ex4.two1", "partition": 0, "offset": 2}
{"timestamp": "2026-10-11T15:12:18.866123+00:00", "kind": "published", "event_id": "8f62f8db-6040-4e33-9725-6d2cae8dcf6a",
"topic": "lab.ex4.two1", "partition": 0, "offset": 3}
{"timestamp": "2026-10-11T15:12:18.866197+00:00", "kind": "published", "event_id": "4238cbf1-d532-484e-8150-1a6e6c7fd019",
```

```
"topic": "lab.ex4.two1", "partition": 0, "offset": 4}
{"timestamp": "2026-10-11T15:12:18.866267+00:00", "kind": "published", "event_id": "c2590858-c544-4c6b-ad35-6635792903f5",
"topic": "lab.ex4.two1", "partition": 0, "offset": 5}
{"timestamp": "2026-10-11T15:12:18.866363+00:00", "kind": "published", "event_id": "4ae34f86-cc1a-47d5-aaa4-0f89f86a2c6a",
"topic": "lab.ex4.two1", "partition": 0, "offset": 6}
{"timestamp": "2026-10-11T15:12:18.866437+00:00", "kind": "published", "event_id": "48dd5490-8a88-4264-8da9-75723c59d647",
"topic": "lab.ex4.two1", "partition": 0, "offset": 7}
{"timestamp": "2026-10-11T15:12:18.866531+00:00", "kind": "published", "event_id": "105732e5-fb27-4091-b92f-4bb700627332",
"topic": "lab.ex4.two1", "partition": 0, "offset": 8}
{"timestamp": "2026-10-11T15:12:18.866613+00:00", "kind": "published", "event_id": "3f98f174-4132-489e-a3e0-132b1635c3d7",
"topic": "lab.ex4.two1", "partition": 0, "offset": 9}
{"timestamp": "2026-10-11T15:12:18.867291+00:00", "kind": "published", "event_id": "013f434d-a723-448c-b0e2-ce871bbadc08",
"topic": "lab.ex4.two1", "partition": 1, "offset": 0}
{"timestamp": "2026-10-11T15:12:18.867448+00:00", "kind": "published", "event_id": "8a1f6870-3665-44b3-a460-1bcb2d60cc6c",
"topic": "lab.ex4.two1", "partition": 1, "offset": 1}
{"timestamp": "2026-10-11T15:12:18.867542+00:00", "kind": "published", "event_id": "40669fac-c08a-4600-a971-29c14338caf5",
"topic": "lab.ex4.two1", "partition": 1, "offset": 2}
{"timestamp": "2026-10-11T15:12:18.867603+00:00", "kind": "published", "event_id": "385fcc5f-51c3-4ce3-bd61-c8355b72291c",
"topic": "lab.ex4.two1", "partition": 1, "offset": 3}
{"timestamp": "2026-10-11T15:12:18.867689+00:00", "kind": "published", "event_id": "1110400d-5d5d-43be-9c09-1b8bf2212002",
"topic": "lab.ex4.two1", "partition": 1, "offset": 4}
{"timestamp": "2026-10-11T15:12:18.867763+00:00", "kind": "published", "event_id": "ee03b4e5-3713-48dc-808b-ea4321e4f76d",
"topic": "lab.ex4.two1", "partition": 1, "offset": 5}
{"timestamp": "2026-10-11T15:12:18.867834+00:00", "kind": "published", "event_id": "1c156fd5-42e2-458f-b3b4-8898f4bdd209",
"topic": "lab.ex4.two1", "partition": 1, "offset": 6}
{"timestamp": "2026-10-11T15:12:18.867907+00:00", "kind": "published", "event_id": "b5e6c50e-dad3-4069-9d71-a5920dfec3a0",
"topic": "lab.ex4.two1", "partition": 1, "offset": 7}
{"timestamp": "2026-10-11T15:12:18.867969+00:00", "kind": "published", "event_id": "9fc82a44-c710-46e8-96f0-c615e7c7e914",
"topic": "lab.ex4.two1", "partition": 1, "offset": 8}
{"timestamp": "2026-10-11T15:12:18.868030+00:00", "kind": "published", "event_id": "3cc18bb9-ebe5-4684-9673-34af43c93f8c",
"topic": "lab.ex4.two1", "partition": 1, "offset": 9}
{"timestamp": "2026-10-11T15:12:18.869082+00:00", "kind": "published", "event_id": "1b211b47-d9b0-4c3d-ad86-4b72368840b4",
"topic": "lab.ex4.two1", "partition": 2, "offset": 0}
{"timestamp": "2026-10-11T15:12:18.869348+00:00", "kind": "published", "event_id": "c6220455-da67-4800-9c90-9b49b6459638",
"topic": "lab.ex4.two1", "partition": 2, "offset": 1}
{"timestamp": "2026-10-11T15:12:18.869465+00:00", "kind": "published", "event_id": "267ef060-4675-4a2f-b158-ecf2b8186eb8",
"topic": "lab.ex4.two1", "partition": 2, "offset": 2}
{"timestamp": "2026-10-11T15:12:18.869537+00:00", "kind": "published", "event_id": "b7ff8fac-1650-4f69-aedf-4e172ed6df67",
"topic": "lab.ex4.two1", "partition": 2, "offset": 3}
{"timestamp": "2026-10-11T15:12:18.869629+00:00", "kind": "published", "event_id": "e17cd8e8-5ca7-4452-84f9-4a704a6d11c4",
"topic": "lab.ex4.two1", "partition": 2, "offset": 4}
{"timestamp": "2026-10-11T15:12:18.869717+00:00", "kind": "published", "event_id": "3b71b045-e07e-4cab-9791-588f2fc871e7",
"topic": "lab.ex4.two1", "partition": 2, "offset": 5}
{"timestamp": "2026-10-11T15:12:18.869803+00:00", "kind": "published", "event_id": "0ebc51da-581c-4e8a-b14c-147b73890834",
"topic": "lab.ex4.two1", "partition": 2, "offset": 6}
{"timestamp": "2026-10-11T15:12:18.869897+00:00", "kind": "published", "event_id": "f33a665c-8e63-4d79-bca1-19b560654c51",
"topic": "lab.ex4.two1", "partition": 2, "offset": 7}
{"timestamp": "2026-10-11T15:12:18.869984+00:00", "kind": "published", "event_id": "2edebb22-9dc5-4744-9f1d-1d8b317e9ba3",
"topic": "lab.ex4.two1", "partition": 2, "offset": 8}
{"timestamp": "2026-10-11T15:12:18.870066+00:00", "kind": "published", "event_id": "391621d8-dadc-469e-933a-4cf00abb24f0",
"topic": "lab.ex4.two1", "partition": 2, "offset": 9}
```


Consumer 1


```
{"timestamp": "2026-10-11T15:11:34.971931+00:00", "kind": "assigned", "consumer": "C1", "group": "lab.ex4.two1",
"partitions": [0, 1, 2, 3]}
{"timestamp": "2026-10-11T15:11:43.978544+00:00", "kind": "revoked", "consumer": "C1", "group": "lab.ex4.two1",
"partitions": [0, 1, 2, 3]}
{"timestamp": "2026-10-11T15:11:43.989796+00:00", "kind": "assigned", "consumer": "C1", "group": "lab.ex4.two1",
"partitions": [0, 1]}
{"timestamp": "2026-10-11T15:12:18.869256+00:00", "kind": "received", "consumer": "C1", "group": "lab.ex4.two1", "topic":
"lab.ex4.two1", "partition": 0, "offset": 0, "key": "order-0", "event_id": "a4ccec2b-9d54-4ff3-abc9-0ede4cccc5c5",
"order_id": "order-0", "sequence": 1, "payload": {"text": "group experiment"}}
{"timestamp": "2026-10-11T15:12:18.869372+00:00", "kind": "completed", "consumer": "C1", "group": "lab.ex4.two1", "topic":
"lab.ex4.two1", "partition": 0, "offset": 0, "key": "order-0", "event_id": "a4ccec2b-9d54-4ff3-abc9-0ede4cccc5c5",
"order_id": "order-0", "sequence": 1, "payload": {"text": "group experiment"}}
{"timestamp": "2026-10-11T15:12:18.892790+00:00", "kind": "committed", "consumer": "C1", "group": "lab.ex4.two1",
"partition": 0, "next_offset": 1}
{"timestamp": "2026-10-11T15:12:18.892933+00:00", "kind": "received", "consumer": "C1", "group": "lab.ex4.two1", "topic":
"lab.ex4.two1", "partition": 0, "offset": 1, "key": "order-4", "event_id": "4541e3e4-1637-4294-9117-6f424c41c6ac",
"order_id": "order-4", "sequence": 1, "payload": {"text": "group experiment"}}
{"timestamp": "2026-10-11T15:12:18.893001+00:00", "kind": "completed", "consumer": "C1", "group": "lab.ex4.two1", "topic":
"lab.ex4.two1", "partition": 0, "offset": 1, "key": "order-4", "event_id": "4541e3e4-1637-4294-9117-6f424c41c6ac",
"order_id": "order-4", "sequence": 1, "payload": {"text": "group experiment"}}
{"timestamp": "2026-10-11T15:12:18.900843+00:00", "kind": "committed", "consumer": "C1", "group": "lab.ex4.two1",
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Consumer 2


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Group of 4

Publisher


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Consumer 1


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Consumer 2


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Consumer 3


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Consumer 4


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Group of 5

Publisher


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Consumer 1


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Consumer 2


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Consumer 3


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Consumer 4


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Consumer 5


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- ☐ Evidence log from the **analytics group** replay showing all 40 records received

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```


☐ Written explanation (3–5 sentences): Why is the 5th consumer idle? Why can the analytics group replay all records that the first group already consumed?

The 5th consumer is idle because a partition can be assigned to only one consumer in a group at a time. The analytics group has its own offsets that start from zero and separate from the first group's offsets. Since Kafka retains the topic records, a new group using `auto.offset.reset=earliest` can start reading from its offset and read the same 40 records again.

Exercise 5 — Kafka: Keyed Ordering

☐ Evidence logs from the **keyed** mode confirming sequence 1→2→3 is preserved for each order


```
consumer,event_id,group,key,kind,next_offset,offset,order_id,partition,partitions,payload,sequence,source_file,timestamp,to
pic,unique_completed
C1,,lab.ex5.keyed1,,assigned,,,,,"[0, 1, 2]",,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:23.920911+00:00,,
C1,,lab.ex5.keyed1,,revoked,,,,,"[0, 1, 2]",,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:29.923690+00:00,,
C1,,lab.ex5.keyed1,,assigned,,,,,"[0, 1]",,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:29.932124+00:00,,
C2,,lab.ex5.keyed1,,assigned,,,,,[2],,,evidence/ex5-keyed-C2.jsonl,2026-10-11T16:04:29.932140+00:00,,
C1,,lab.ex5.keyed1,,revoked,,,,,"[0, 1]",,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:35.935533+00:00,,
C2,,lab.ex5.keyed1,,revoked,,,,,[2],,,evidence/ex5-keyed-C2.jsonl,2026-10-11T16:04:35.935534+00:00,,
C2,,lab.ex5.keyed1,,assigned,,,,,[1],,,evidence/ex5-keyed-C2.jsonl,2026-10-11T16:04:35.943934+00:00,,
C1,,lab.ex5.keyed1,,assigned,,,,,[0],,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:35.943935+00:00,,
C3,,lab.ex5.keyed1,,assigned,,,,,[2],,,evidence/ex5-keyed-C3.jsonl,2026-10-11T16:04:35.943949+00:00,,
C2,3453ec1d-5668-40c1-92cf-d753a3fe0c52,lab.ex5.keyed1,order-123,received,,0,order-123,1,,"{""action"":
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C1,a35a71c4-357b-441b-8e5c-ab03960ce070,lab.ex5.keyed1,order-789,received,,0,order-789,0,,"{""action"":
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```

```
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C2,,lab.ex5.keyed1,,committed,1,,,1,,,,evidence/ex5-keyed-C2.jsonl,2026-10-11T16:04:47.601532+00:00,,
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C3,,lab.ex5.keyed1,,revoked,,,,,[2],,,evidence/ex5-keyed-C3.jsonl,2026-10-11T16:04:58.723993+00:00,,
C3,,,,summary,,,,,,,,evidence/ex5-keyed-C3.jsonl,2026-10-11T16:04:58.725330+00:00,,3
C2,,,,interrupted,,,,,,,,evidence/ex5-keyed-C2.jsonl,2026-10-11T16:04:59.724496+00:00,,
C2,,lab.ex5.keyed1,,revoked,,,,,[1],,,evidence/ex5-keyed-C2.jsonl,2026-10-11T16:04:59.724681+00:00,,
C2,,,,summary,,,,,,,,evidence/ex5-keyed-C2.jsonl,2026-10-11T16:04:59.726015+00:00,,3
C1,,lab.ex5.keyed1,,revoked,,,,,[0],,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:59.954852+00:00,,
C1,,lab.ex5.keyed1,,assigned,,,,,"[0, 1, 2]",,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:59.962925+00:00,,
C1,,,,interrupted,,,,,,,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:05:01.125278+00:00,,
C1,,lab.ex5.keyed1,,revoked,,,,,"[0, 1, 2]",,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:05:01.125519+00:00,,
C1,,,,summary,,,,,,,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:05:01.133287+00:00,,3
```

- Evidence logs from the **broken** mode showing out-of-order completion

```
consumer,event_id,group,key,kind,next_offset,offset,order_id,partition,partitions,payload,sequence,source_file,timestamp,to
pic,unique_completed
C1,,lab.ex5.keyed1,,assigned,,,,,"[0, 1, 2]",,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:23.920911+00:00,,
C1,,lab.ex5.keyed1,,revoked,,,,,"[0, 1, 2]",,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:29.923690+00:00,,
C1,,lab.ex5.keyed1,,assigned,,,,,"[0, 1]",,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:29.932124+00:00,,
C2,,lab.ex5.keyed1,,assigned,,,,,[2],,,evidence/ex5-keyed-C2.jsonl,2026-10-11T16:04:29.932140+00:00,,
C1,,lab.ex5.keyed1,,revoked,,,,,"[0, 1]",,,evidence/ex5-keyed-C1.jsonl,2026-10-11T16:04:35.935533+00:00,,
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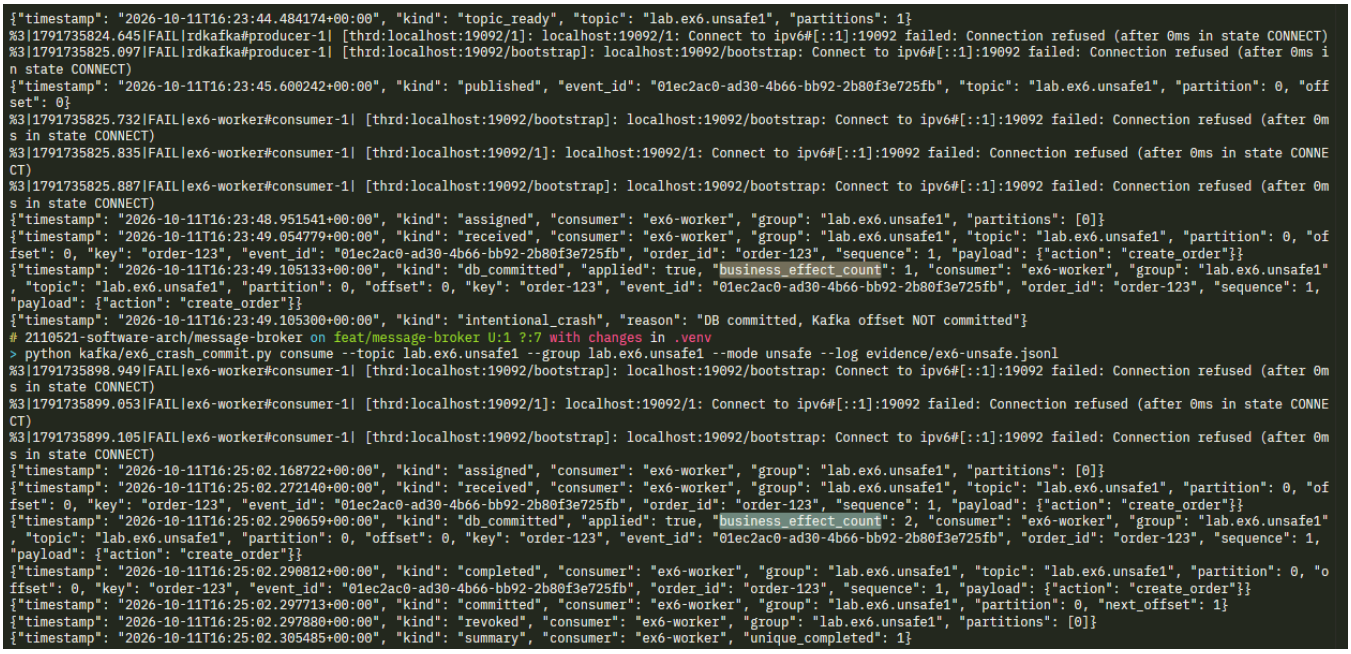
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```

- ☐ Written explanation (3–5 sentences): What causes the difference between the two modes? Who is responsible for ensuring business-level ordering — the producer, the broker, or the consumer?

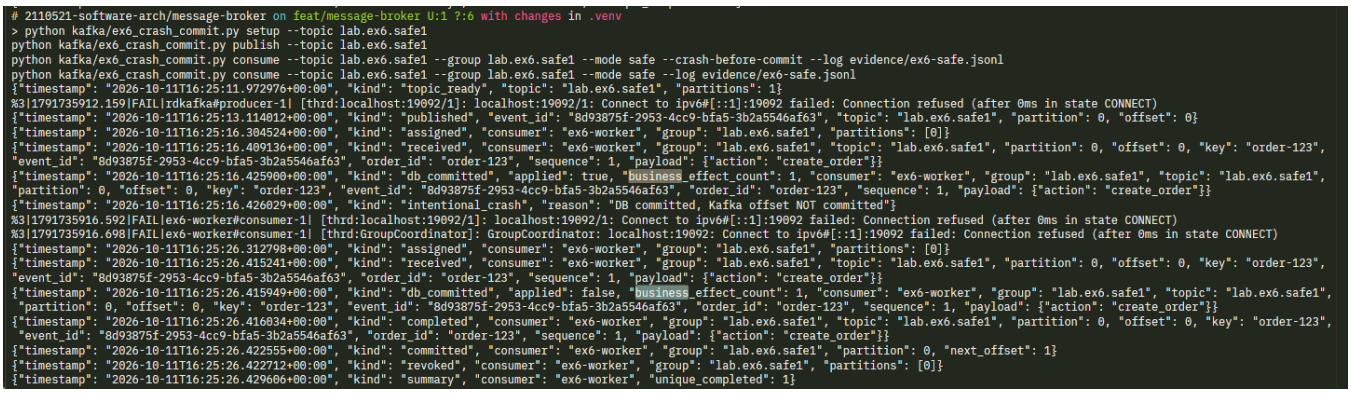
In keyed mode, events for the same order use the same key, so Kafka's partitioner sends them to one partition and preserves their order there. In broken mode, the events are sent to different partitions, where separate consumers process them concurrently. To ensure ordering, the producer is responsible for choosing a key or partitioning strategy that keeps business-related events together. Kafka preserves order within a partition, but not across partitions, and consumers should avoid processing a partition's records in a way that breaks the order.

Exercise 6 — Kafka: Crash Before Offset Commit

- ☐ Terminal output from the **unsafe** mode — crash run and restart, showing `applied` and `business_effect_count`



- ☐ Terminal output from the **safe** mode — crash run and restart, showing `applied` and `business_effect_count`



- ☐ Written explanation (3–5 sentences): Compare the Kafka offset commit mechanism in this exercise with the RabbitMQ ACK mechanism in Exercise 1. What is the same? What is different?

Both RabbitMQ ACK and Kafka offset commits record processing progress, so a crash before progress recording can cause the message to be processed again. In unsafe mode, that replay applies the effect a second time. In safe mode, event-ID deduplication keeps the effect count at one. RabbitMQ acknowledges individual deliveries and requeues an unacknowledged message after a consumer failure. Kafka records the next offset for a consumer group and partition, so a restart resumes from its last committed position.

Overall Submission Checklist

Combine all of the above into **one PDF or Markdown file** and submit before the deadline.

- ☐ Exercise 1 — unsafe and safe terminal output + written explanation
- ☐ Exercise 2 — durable and temporary terminal output + written explanation
- ☐ Exercise 3 — 3-worker and 1-worker terminal output + written explanation
- ☐ Exercise 4 — partition assignment logs (groups of 2, 4, 5) + analytics replay log + written explanation
- ☐ Exercise 5 — keyed and broken evidence logs + written explanation
- ☐ Exercise 6 — unsafe and safe terminal output + written explanation
- ☐ Overall Reflection (5–8 sentences): Which failure mode surprised you most, and why? What design principle would you apply in your own system to handle message redelivery safely?

Late subscriber surprised me the most, because it introduces me to difference between broker and topic queue in RabbitMQ, and that the queue in RabbitMQ waits until consumer wakes up to send the message. I would apply the principle about making the consumer to be smart and be able to reconcile itself regardless of how broken the data is.

Concepts Summary

The table below shows which core concepts each exercise covers.

Concept	Ex 1	Ex 2	Ex 3	Ex 4	Ex 5	Ex 6
At-least-once delivery	✅					✅
Idempotency / deduplication	✅					✅
Acknowledgement / offset commit	✅					✅
Queue durability & binding time		✅				
Fanout / pub-sub routing		✅				
Work queue / competing consumers			✅			
Concurrency & completion order			✅			
Consumer groups & partition assignment				✅		
Independent group replay				✅		
Key-based partitioning & ordering					✅	
Cross-partition ordering failure					✅	
Producer idempotence					✅	
Crash recovery & progress tracking	✅					✅
RabbitMQ vs Kafka comparison						✅